60 IT 403 - Web Technology

STUDYMATE AI

AI Powered Student Productivity and Learning Assistance Platform

Sangamithra V K (240373720522113)

Sandhiya S (240373720522109)

Department of Information Technology

K.S.RANGASAMY COLLEGE OF TECHNOLOGY

(Autonomous)

Abstract

StudyMate AI is an AI-powered student productivity and learning assistance platform developed to support students in their academic activities. The system provides features such as AI text summarization, OCR text extraction, audio-to-text conversion, AI chatbot assistance, quiz generation, study planning, and doubt solving. It is developed using HTML, CSS, JavaScript, Python Flask, MySQL, and AI technologies. The platform helps students save time, improve learning efficiency, and manage academic tasks through a single user-friendly application.

Introduction

- Students often face difficulties in managing study materials and academic activities.
- Artificial Intelligence helps improve learning efficiency and productivity.
- StudyMate AI is a smart web-based learning assistance platform for students.
- The system provides features such as AI Summarizer, OCR, Audio-to-Text, Chatbot, Quiz Generator, and Study Planner.
- It is developed using HTML, CSS, JavaScript, Python Flask, MySQL, and AI technologies.

Problem Statement

Existing Problems

- Time-consuming note preparation
- Difficulty in summarizing lengthy content
- Managing schedules manually
- Lack of instant doubt clarification
- Multiple applications required for different tasks

Proposed Solution

- StudyMate AI provides all educational assistance tools in one platform using AI technologies.

Objectives

- Provide AI-powered text summarization
- Convert images to text using OCR
- Convert audio lectures to notes
- Generate quizzes automatically
- Provide AI chatbot support
- Create study plans and reminders
- Generate professional resumes
- Solve academic doubts instantly

Technologies Used

Frontend

- HTML5
- CSS3
- Bootstrap
- JavaScript

Backend

- Python Flask

Database

- MySQL

Modules

Authentication Module

- Login
- Registration

AI Learning Modules

- Summarizer
- OCR
- Audio To Text
- Chatbot
- Quiz Generator

Productivity Modules

- Study Planner
- Reminders
- Knowledge Base
- Resume Builder
- PDF Summarizer

Advanced Features

AI Quiz Generator

- Generates MCQ questions automatically

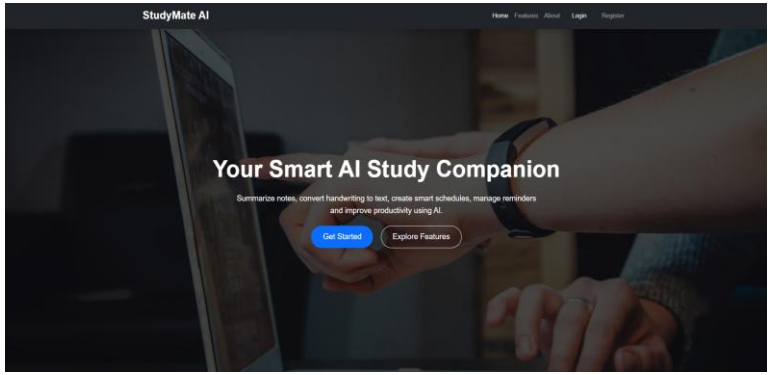
AI PDF Summarizer

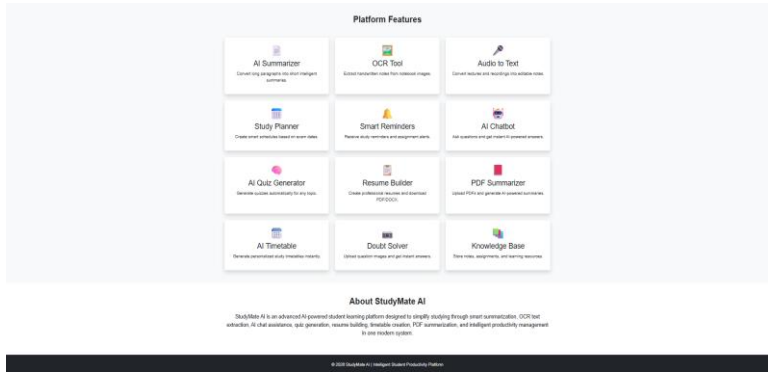
- Summarizes uploaded PDFs

AI Doubt Solver

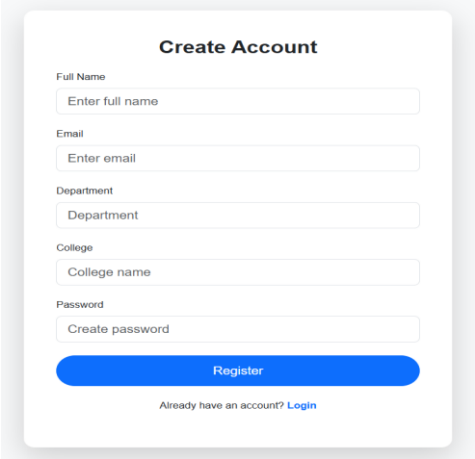
- Upload question image
- OCR extraction
- AI-generated answer

Screenshots

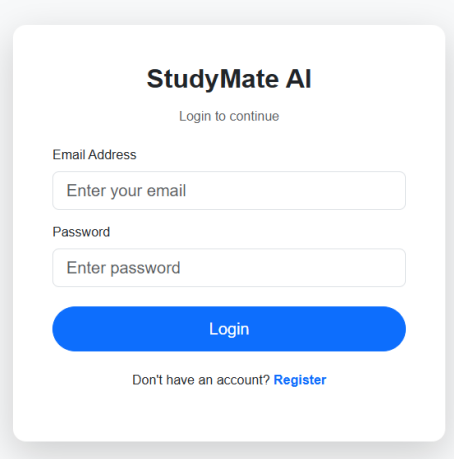




Home Page

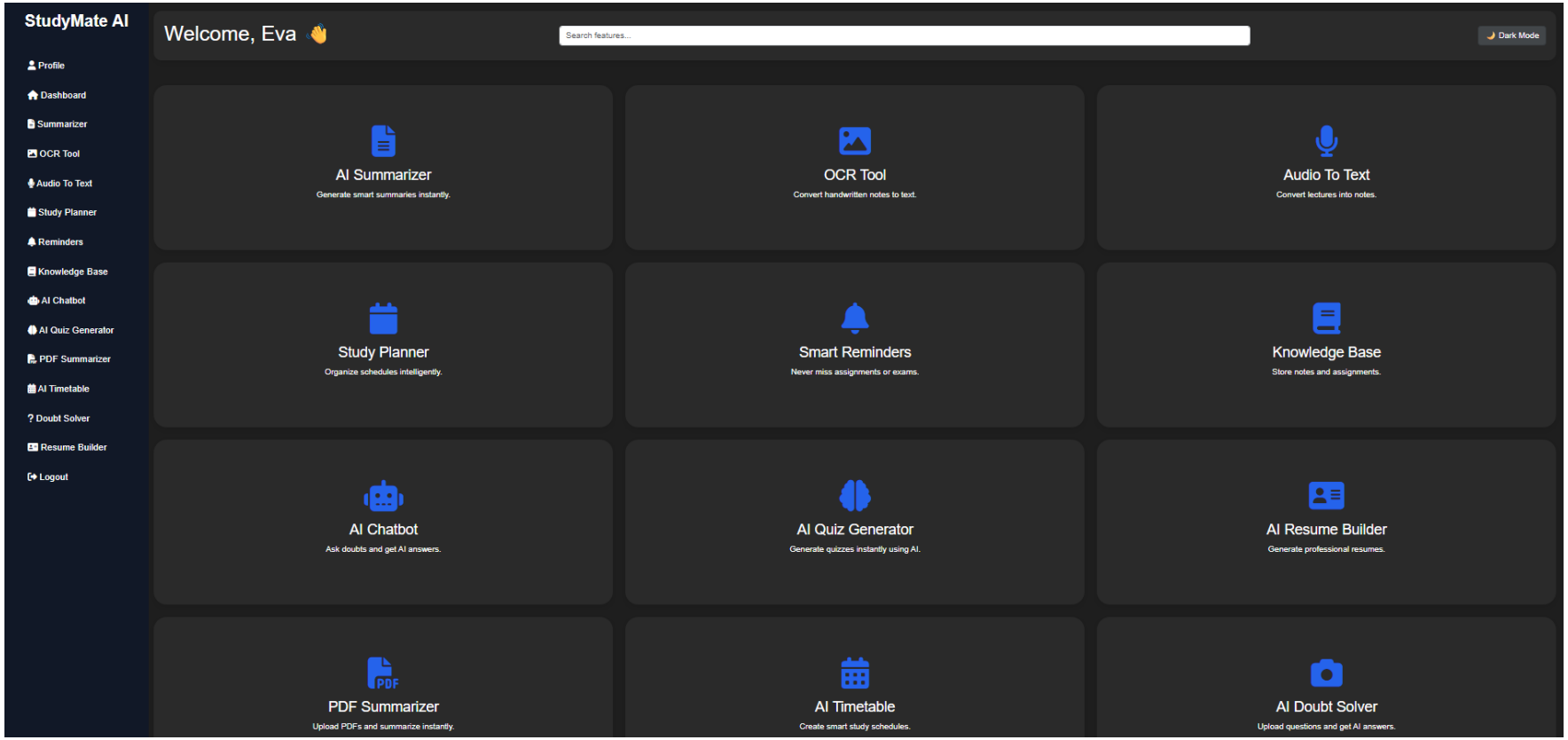


Register



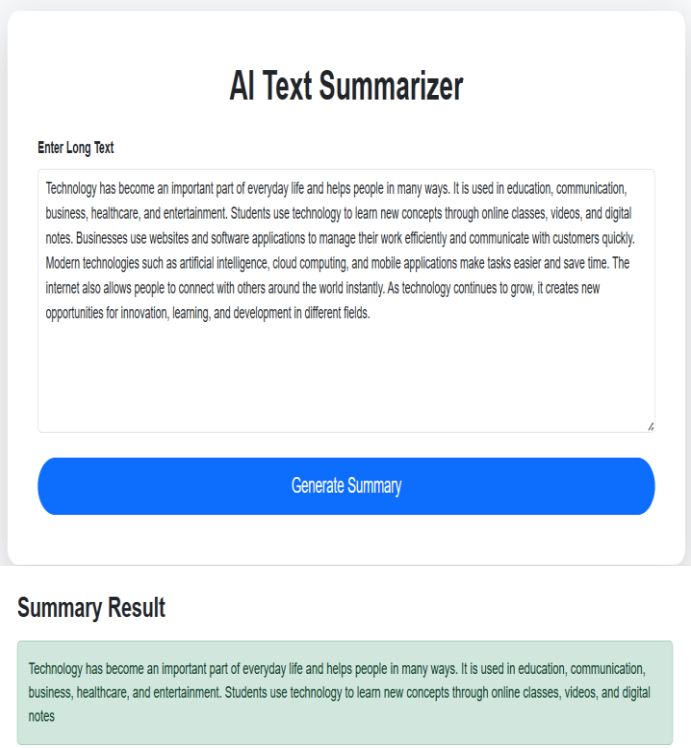
Login

Screenshots

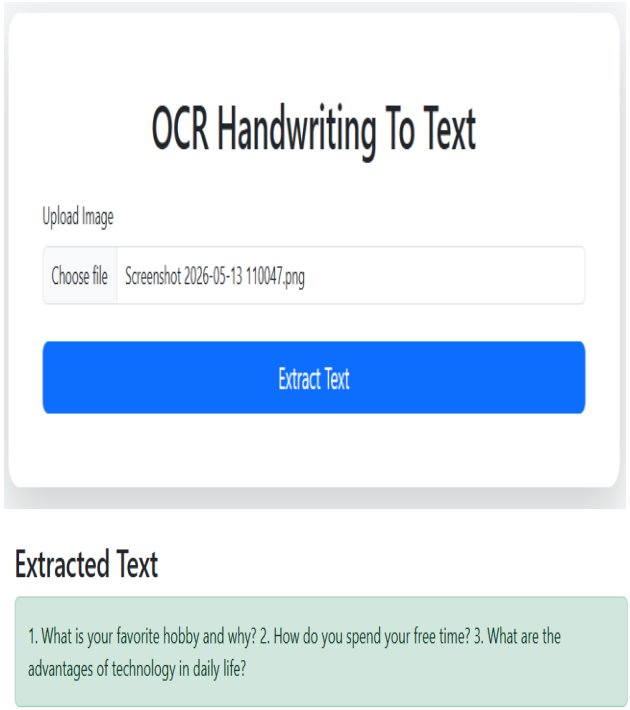


Dashboard

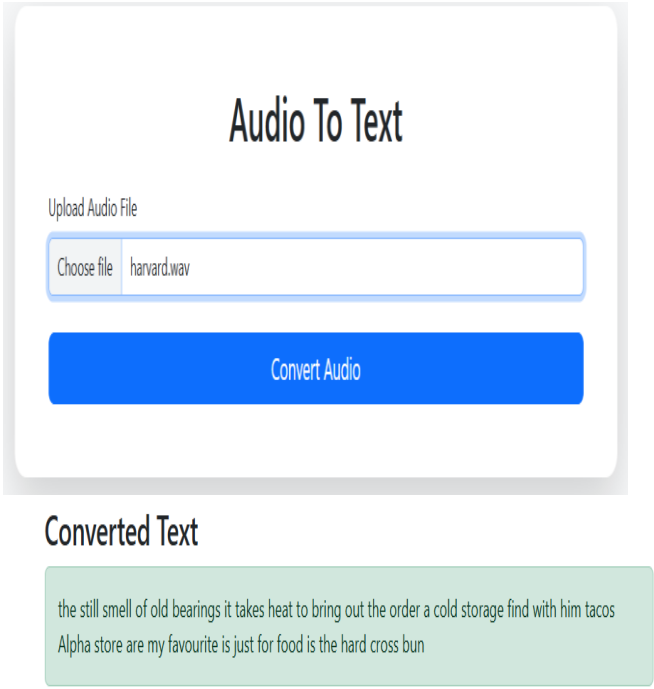
Screenshots



AI summarizer

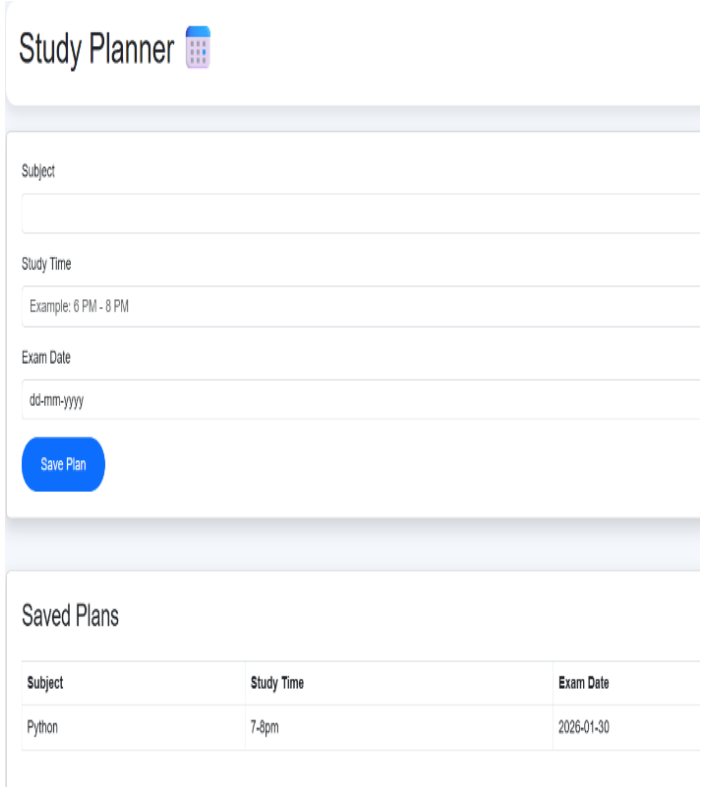


OCR Tool

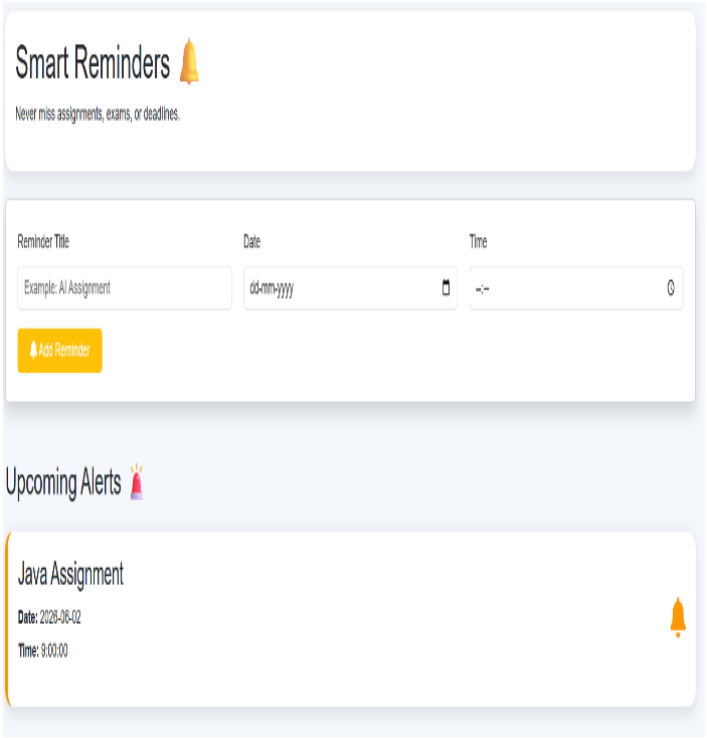


Audio to Text

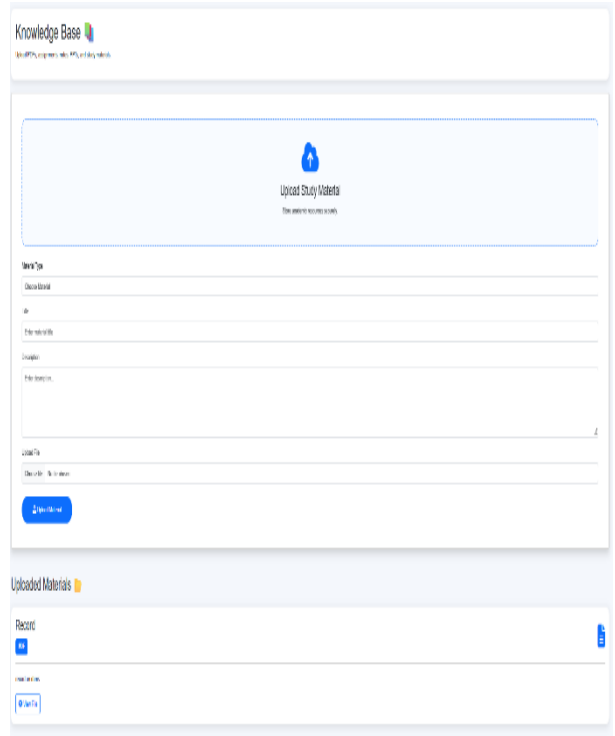
Screenshots



Study Planner

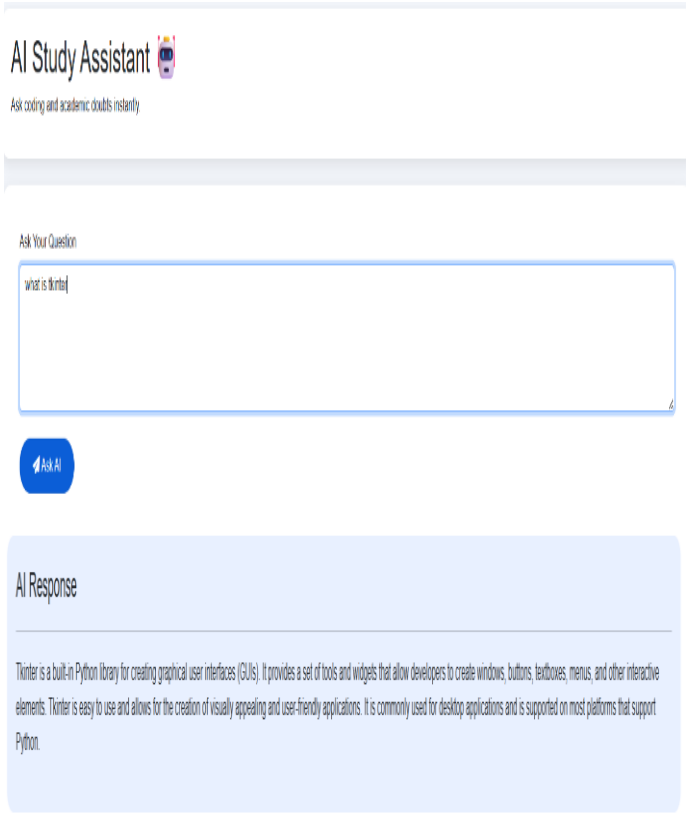


Smart Remainder

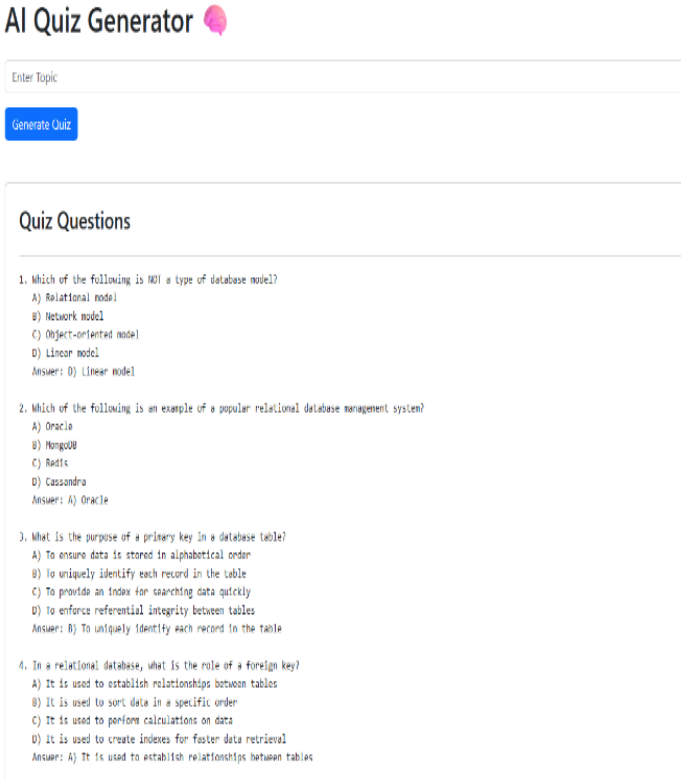


Knowledge Base

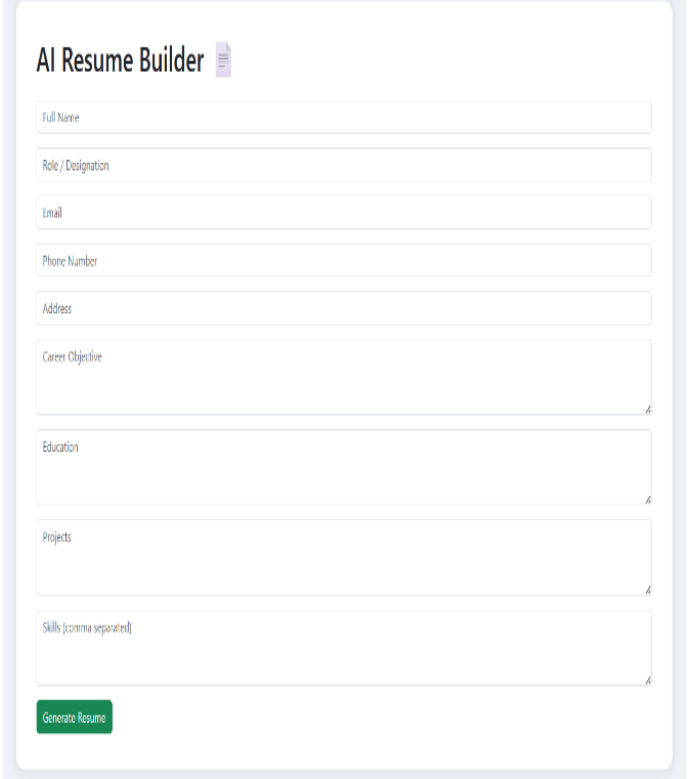
Screenshots



AI Chatbot

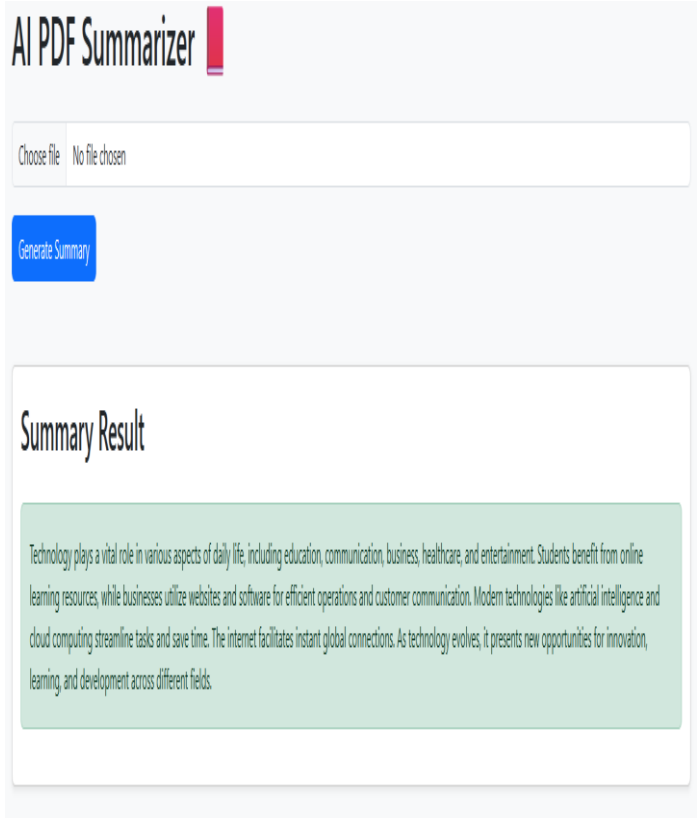


AI Quiz Generator

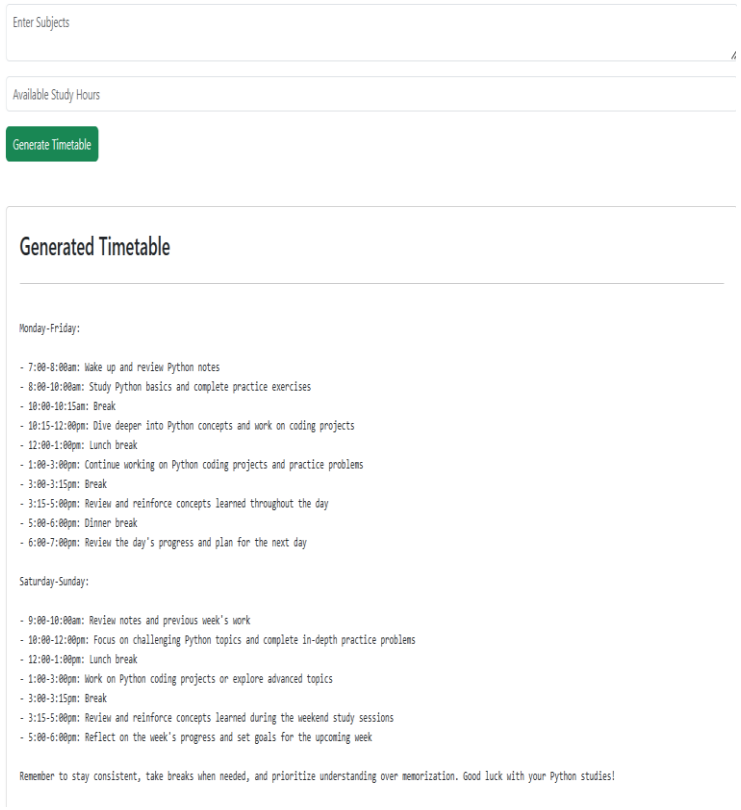


AI Resume Builder

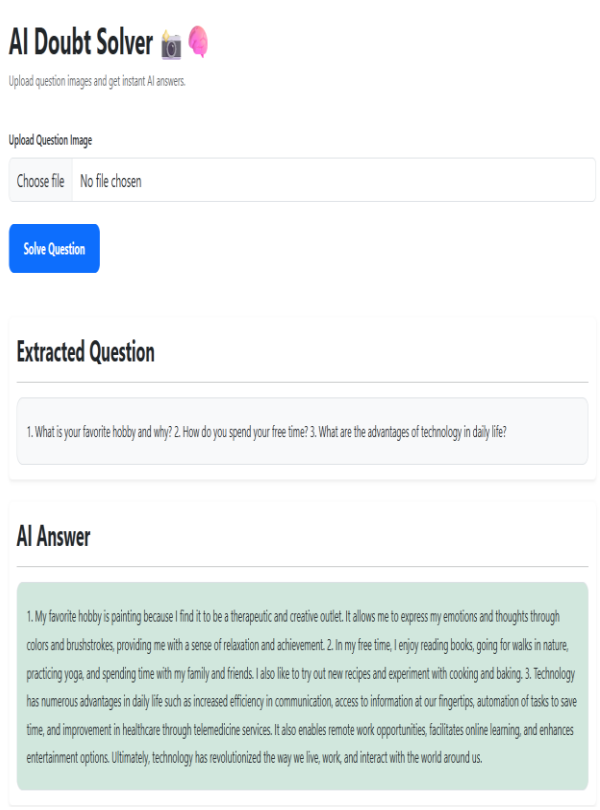
Screenshots



PDF Summarizer



AI Timetable Generator



AI Doubt Solver

Advantages

- Improves student productivity
- Saves study time
- Centralized learning platform
- Easy-to-use interface
- AI-powered academic assistance
- Supports efficient learning

Future Enhancements

- Voice Assistant Integration
- Multi-language Support
- AI-Based Performance Analytics
- Online Collaboration Features
- Mobile Application Development
- Cloud Storage Integration

Conclusion

- StudyMate AI successfully integrates multiple AI-powered educational tools into a single platform.
- It improves student productivity and academic organization
- Resume generation supports placement preparation
- The system helps students improve productivity, manage academic activities efficiently.
- StudyMate AI promotes smart digital education systems

Thank You



K.S.RANGASAMY COLLEGE OF TECHNOLOGY

(Autonomous)

Tiruchengode - 637 215. Namakkal Dt. Tamil Nadu. India

www.ksrct.ac.in

/ksrct1994